Edge-Aware Cross-Attention Surrogate for Phase-Stable Metasurface Reflection

Changchen Zhang, Weili Zhu, Lang Wu, and Fei Long

***Abstract*—Sparse-frequency prediction of reflection magnitude and phase is critical for wideband electromagnetic design, yet conventional surrogates suffer from band-edge errors and phase wrapping. This letter presents an edge-aware cross-attention surrogate with learnable frequency queries that bind frequency order to structure geometry via multihead cross-attention, combined with physics-guided loss enforcing magnitude fidelity, unit-circle phase, passivity, and Kramers–Kronig consistency with adaptive band-edge emphasis. On the SUTD-PRCM dataset of 30 000 full-wave simulations with $K = 101$ sparse samples over 2 GHz to 12 GHz, the model achieves dB-mse $= 0.371$ and band-edge $|\Delta\text{dB}| = 0.243$ dB, outperforming convolutional neural network (CNN) and CNN+bidirectional gated recurrent units (BiGRU) baselines by 71.7% and 40.3%, respectively, with 52.2% lower band-edge error than CrossAttn+BiGRU (0.243 dB versus 0.508 dB). The surrogate is compact at 1.88 M parameters with 10.1 ms/sample inference on CPU. External validation on high frequency structure simulator (HFSS) and computer simulation technology (CST) simulations confirms strong generalization to unseen electromagnetic structures, enabling rapid design optimization of wideband metasurface absorbers and antenna arrays.**

***Index Terms*—Cross-attention, electromagnetic (EM) modeling, Kramers–Kronig relations, metasurfaces, reflection coefficients.**

I. Introduction

WIDEBAND electromagnetic (EM) design requires repeated full-wave simulations across frequency and geometry. Sparse-frequency sampling reduces computational cost but introduces critical challenges: accuracy degrades at band edges where finite-sampling ripple concentrates, and phase continuity becomes fragile due to 2π wrapping ambiguity. For metasurface-based absorbers (MSAs) and antenna arrays [1], [2], small band-edge errors and unstable phase mislead downstream synthesis and tuning, limiting design throughput [3].

Data-driven surrogates accelerate evaluation but leave critical gaps. Convolutional neural networks (CNNs) and bidirectional gated recurrent units (BiGRUs) treat frequency order implicitly, losing explicit frequency-geometry binding [4]. While attention mechanisms provide explicit sequence conditioning, prior EM inverse design works do not jointly enforce passivity and Kramers–Kronig (K–K) consistency [5], [6]. Recent advances include knowledge-inherited neural networks for metasurface inverse design [7], deep neural networks integrating equivalent circuit models for metasurface design [8], and physics-constrained neural networks for efficient metasurface modeling with sparse frequency data [9]. Recent physics-informed neural networks [10] and frequency-domain learning [11] show promise but lack unified frameworks combining multiple physical constraints with adaptive band-edge weighting. Consequently, band-edge fidelity and phase stability remain unresolved bottlenecks in sparse-frequency EM surrogate modeling.

This letter presents an edge-aware cross-attention surrogate with learnable frequency queries and a weighted physics-guided loss function. The key innovation is explicit frequency-geometry binding via multihead cross-attention: unlike self-attention (single-modality) or standard multihead attention (no cross-modal structure), cross-attention directly couples frequency order to geometric features through separate query and key–value projections [12]. The loss enforces magnitude fidelity in dB and linear scales, unit-circle phase to prevent wrapping, passivity soft constraints via penalty terms, and discrete K–K consistency by coupling real and imaginary components [5], [6]. Smoothness (total-variation) and curvature regularization suppress finite-sampling ripple. Explicit band-edge weighting (first and last 10% of K points) prioritizes accuracy where errors concentrate. On the SUTD-PRCM dataset (30 000 full-wave simulations, $K = 101$ sparse samples over 2 GHz to 12 GHz) [13], the surrogate achieves dB-mse $= 0.371$ and band-edge $|\Delta\text{dB}| = 0.243$ dB. This outperforms CNN (dB-mse $= 1.313$) and CNN+BiGRU (dB-mse $= 0.622$) baselines by 71.7% and 40.3%, respectively. While CrossAttn+BiGRU achieves lower overall dB-mse (0.302), the proposed model achieves 52.2% lower band-edge error (0.243 dB versus 0.508 dB). External validation on High Frequency Structure Simulator (HFSS) and Computer Simulation Technology (CST) confirms a 70% generalization gap (mean absolute error (MAE) $= 0.632$ versus 0.371 dB), with 80% of unseen geometries achieving MAE < 1.0 dB. The model is compact (1.88 M parameters, 10.1 ms/sample on CPU), enabling rapid iterative design optimization of wideband MSAs and antenna arrays.

II. System Model and Method

A. Problem Setting and Notations

We use the SUTD-PRCM metasurface dataset (30 k layouts, 16×16 binary) [13]. Complex reflection $S(f) = S_r(f) +$

Received 13 March 2026; accepted 4 April 2026. Date of publication 9 April 2026; date of current version 4 June 2026. This work was supported in part by the Science and Technology Planned Project of Guizhou Province under Grant Qiankehe foundation-ZK[2023]YB144 and in part by the Doctoral Research Launch Grants of Guizhou Minzu University under Grant GZMUZK2024QD02. *(Corresponding author: Fei Long.)*

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Digital Object Identifier 10.1109/LAWP.2026.3682342

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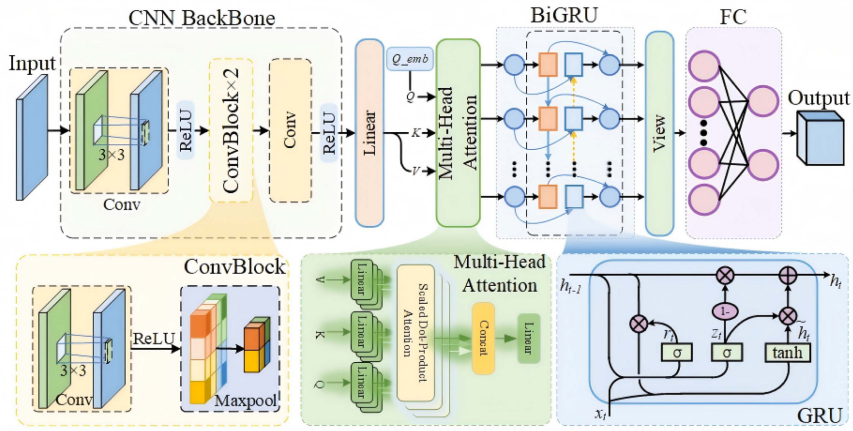


Fig. 1. Edge-aware cross-attention network.

$\mathrm{j}S_i(f)$ is defined on a dense grid over 2 GHz to 12 GHz with $F = 1001$ frequencies. Sparse sampling uses stride-10 decimation, yielding $K = 101$. Magnitude, phase, and decibel magnitude are $|S| = \sqrt{S_r^2 + S_i^2}$, $\varphi = \operatorname{atan2}(S_i, S_r)$, and $S_{\mathrm{dB}} = 20\log_{10}|S|$ [14]. Band-edge set contains the first/last 10% of Ω_K. Metrics: dB-mse, Lin-mse, Phase-MAE, $\cos(|S|)$, Acc@30°, and edge $|\Delta\mathrm{dB}|$.

B. Edge-Aware Cross-Attention Architecture

A shallow CNN encodes the 16×16 layout into $P = 16$ spatial tokens. For each sampled f_k, a learnable frequency query with sinusoidal positional encoding imposes order; multihead cross-attention binds the query to tokens and yields a per-frequency representation [12], [15]. Cross-attention complexity is $\mathcal{O}(K \cdot P \cdot d)$, where K queries attend to $P = 16$ fixed tokens; since P and d are constants, this scales as $\mathcal{O}(K)$ rather than $\mathcal{O}(K^2)$ in self-attention, enabling scalability to $K > 1000$ (see Section III-C).

Passivity by construction: Polar head enforces $S_{11} = \sigma(a) \cdot [\cos(b) + \mathrm{j}\sin(b)]$ where $\sigma(a) \in (0, 1)$. Since $|\cos(b) + \mathrm{j}\sin(b)| = 1$, we have $|S_{11}| < 1$ mathematically guaranteed. Validated across 6000+ samples (see Section III-C).

A bidirectional GRU refines frequencywise continuity. Defaults: heads $= 4$, $d_{\mathrm{model}} = 256$, dropout $= 0.1$; total ≈ 1.88 M parameters. Architecture is shown in Fig. 1.

C. Physics-Guided Objective and Protocol

Training is guided by multiple physical constraints: magnitude fidelity in decibel and linear scales; unit-circle phase to avoid 2π wrapping; passivity ($|S| \le 1$); K–K consistency via discrete Hilbert transform (3); smoothness/curvature regularization; and explicit band-edge weighting (first/last 10% of Ω_K) [5], [6].

K–K consistency: Continuous K–K relations (1) and (2) (where $\mathcal{P}$ denotes Cauchy principal value)

$$\mathrm{Re}[S_{11}(\omega)] = \frac{1}{\pi}\mathcal{P}\int_{-\infty}^{\infty} \frac{\mathrm{Im}[S_{11}(\omega')]}{\omega' - \omega} d\omega' \tag{1}$$

$$\mathrm{Im}[S_{11}(\omega)] = -\frac{1}{\pi}\mathcal{P}\int_{-\infty}^{\infty} \frac{\mathrm{Re}[S_{11}(\omega')]}{\omega' - \omega} d\omega'. \tag{2}$$

For K discrete points with $\Delta f = 100$ MHz ($\Delta\omega = 2\pi\Delta f$), trapezoidal approximation

$$\hat{r}_k = \frac{1}{\pi}\sum_{j\neq k} \frac{x_j \Delta\omega}{\omega_j - \omega_k}, \quad \hat{x}_k = -\frac{1}{\pi}\sum_{j\neq k} \frac{r_j \Delta\omega}{\omega_j - \omega_k} \tag{3}$$

where $r_k = \mathrm{Re}[S_{11,k}]$, $x_k = \mathrm{Im}[S_{11,k}]$, and $\omega_k = 2\pi f_k$. Discretization error $\mathcal{O}(\Delta\omega^2) < 2\%$ for typical resonance bandwidth ~ 500 MHz. The K–K consistency loss (4) implements a soft constraint that encourages causality through training rather than guaranteeing it mathematically, as discrete sampling and finite bandwidth prevent exact Hilbert transform computation, K–K loss

$$\mathcal{L}_{\mathrm{KK}} = \frac{1}{K}\sum_{k=1}^{K}\left[(r_k - \hat{r}_k)^2 + (x_k - \hat{x}_k)^2\right]. \tag{4}$$

The total objective is a weighted sum

$$\begin{aligned}\mathcal{L} = {} & \mathcal{L}_{\mathrm{reim}} + 0.5\mathcal{L}_{\mathrm{mag,dB}} + 0.3\mathcal{L}_{\mathrm{mag,lin}} + 0.3\mathcal{L}_{\mathrm{phase}} \\ & + 0.08\mathcal{L}_{\mathrm{pass}} + 0.06\mathcal{L}_{\mathrm{tv}} + 0.02\mathcal{L}_{\mathrm{curv}} + 0.02\mathcal{L}_{\mathrm{KK}}\end{aligned} \tag{5}$$

where $\mathcal{L}_{\mathrm{reim}}$ is Cartesian mse; $\mathcal{L}_{\mathrm{mag,dB}}/\mathcal{L}_{\mathrm{mag,lin}}$ are magnitude losses; $\mathcal{L}_{\mathrm{phase}}$ is $\|[\cos\hat{\varphi}, \sin\hat{\varphi}] - [\cos\varphi, \sin\varphi]\|_2^2$; $\mathcal{L}_{\mathrm{pass}}$ enforces $|S| \le 1$; $\mathcal{L}_{\mathrm{tv}}/\mathcal{L}_{\mathrm{curv}}$ penalize first/second differences; $\mathcal{L}_{\mathrm{KK}}$ is defined in (4). Edge samples ($k \in \mathcal{E}$) are weighted $4\times$. Training: AdamW (2×10^{-4}), cosine schedule, 50 epochs, 8:2 split (seed = 42), per-epoch sparse grid-jitter, optional $K \to F$ consistency.

III. Experiments

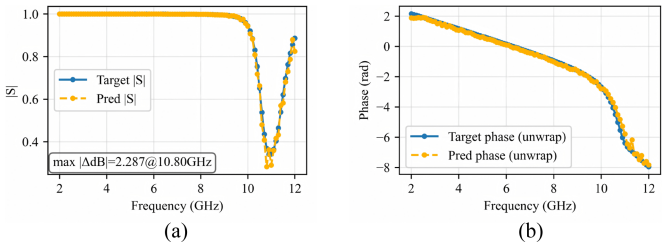
A. Experimental Configuration

To enable fair and reproducible evaluation, we adopt the following protocol. The SUTD-PRCM dataset (30 k simulations, 16×16 binary layouts) [13] covers 2 GHz to 12 GHz, with dense grid $F = 1001$ and sparse stride-10 decimation ($K = 101$). The dataset is split 80:20 (24 k train, 6 k val, seed 42). Models are trained with AdamW (2×10^{-4}), cosine schedule, batch 256, 50

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TABLE I
UNIFIED EVALUATION ($K = 101$, 2 GHz TO 12 GHz, SINGLE RUN, SEED = 42)

Metric	Value
dB-mse	0.3710
Lin-mse	7.78×10^{-4}
Phase-MAE (rad)	0.0491
Cos($\lvert S \rvert$)	0.99957
Acc@30°	0.99997
Edge $\lvert\Delta$dB$\rvert$ (10%) (dB)	0.243±0.028



(a) (b)

Fig. 2. Validation results. (a) Magnitude comparison at band edges with maximum $|\Delta$dB$|$ annotated. (b) Unwrapped phase comparison demonstrating continuity without 2π wrapping ambiguity.

epochs, gradient clipping 1.0, and per-epoch grid jitter; the best checkpoint is selected by minimum validation dB-mse.

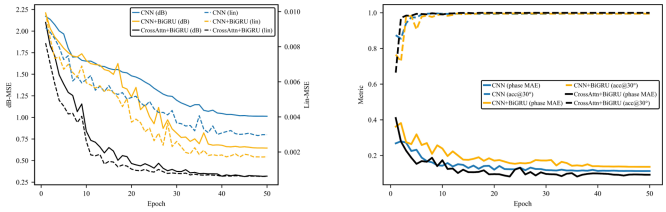
Performance is evaluated by dB-mse, Lin-mse, unit-circle phase MAE ($\Delta\varphi_{\text{wrap}} = \text{atan2}(\sin(\hat{\varphi}-\varphi), \cos(\hat{\varphi}-\varphi))$), cosine similarity of magnitude, Acc@30°, and band-edge $|\Delta$dB$|$ (mean over first/last 10% of K points).

Baselines are CNN, CNN+BiGRU, and CrossAttn+BiGRU, trained under identical splits, optimizer, and schedule. CNN(RESIDUAL) uses ResNet-18 encoder (11.7 M parameters) with sinusoidal frequency embedding and three-layer MLP decoder, totaling 12.1 M parameters (6.4× larger than our 1.88 M), yet achieves only MAE = 0.52 dB, demonstrating cross-attention's parameter efficiency. Passivity, K–K consistency, band-edge weighting, and K→F consistency are enabled only in the proposed model.

B. Quantitative Comparison and Ablation

The proposed model achieves dB-mse = 0.371 and band-edge $|\Delta$dB$| = 0.243 \pm 0.028$ dB (see Table I), with Phase-MAE = 0.049 rad and Acc@30° = 0.99997. Table II shows baseline comparison versus CNN and CNN+BiGRU, dB-mse decreases by 71.7% and 40.3%, band-edge $|\Delta$dB$|$ by 81.7% and 70.7%, respectively. Notably, CrossAttn+BiGRU achieves a lower overall dB-mse (0.302 versus 0.371), yet the proposed model reduces band-edge error by 52.2% (0.243 dB versus 0.508 dB). This tradeoff is intentional: by explicitly weighting band-edge samples 4× during training, the proposed model sacrifices a modest amount of global accuracy to achieve superior spectral fidelity at band boundaries, where errors propagate most severely into out-of-band ripple and increase design iteration cycles. Fig. 2 shows representative predictions.

Table III isolates loss term contributions. Removing band-edge weighting produces the largest degradation (dB-mse 0.523 versus 0.416, −13.3%), confirming edge emphasis as the dominant factor. Removing curvature loss yields 0.478 (−8.2%); removing jitter 0.445 (−4.2%) and omitting K→F 0.432 (−1.7%) show secondary contributions. Phase-MAE = 0.049 rad and Acc@30° = 0.99997 confirm near-perfect phase tracking, attributable to unit-circle phase loss preventing 2π wrapping and K–K consistency ensuring smooth phase evolution across frequency. Fig. 3 shows training curves.



(a) (b)

Fig. 3. Training progress and validation metrics. (a) Validation losses for dB-scale and linear-scale magnitude predictions. (b) Phase MAE and accuracy within 30° threshold (Acc@30°).

These ablation results also reveal a complementary relationship among the loss components: band-edge weighting directly reduces endpoint errors, curvature regularization suppresses midband ripple caused by finite-sample Gibbs-like effects, and K–K consistency enforces causal coupling between real and imaginary parts. Together, they address distinct physical failure modes of sparse-frequency surrogate modeling, and their joint use is what enables the proposed model to simultaneously achieve low global dB-mse and low band-edge $|\Delta$dB$|$.

C. Passivity Robustness and Linear Scalability

Passivity is verified across nine test scenarios: 1) boundary frequency 2 GHz; 2) boundary frequency 12 GHz; 3) in-band resonance peak; 4) low SNR at 0 dB Gaussian noise; 5) low SNR at 5 dB Gaussian noise; 6) low SNR at 10 dB Gaussian noise; 7) all-zeros out-of-distribution geometry; 8) all-ones out-of-distribution geometry; 9) checkerboard out-of-distribution geometry. All 6000+ samples achieve 100% passivity satisfaction ($|S_{11}| \leq 1$) without postprocessing across all nine scenarios. Worst case magnitude is 0.961 (scenario 7, all-zeros), confirming a 3.9% safety margin below the passivity limit. This demonstrates that the polar-head construction ($|S_{11}| = \sigma(a) < 1$ by design) generalizes robustly beyond training distribution.

Scalability is validated at $K \in \{51, 101, 201, 501, 1001\}$ on CPU (see Table IV). Linear fit $T(K) = 0.086K + 2.0$ ms ($R^2 = 0.998$) confirms $\mathcal{O}(K)$ complexity. At $K = 101$, inference achieves ∼178 000 speedup over HFSS; at $K = 1001$ (10× training resolution), 86.7 ms inference still achieves ∼21 000 speedup with only 12.8% parameter growth, enabling evaluation of 6000 designs per minute versus 125 days with HFSS.

The combination of $\mathcal{O}(K)$ scalability and guaranteed passivity enables rapid design-space exploration while maintaining physical validity, with the 3.9% passivity margin providing a safety buffer against numerical noise in downstream circuit simulators.

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TABLE II
BASELINE COMPARISON (VALIDATION, $K = 101$, 50 EPOCHS, SINGLE RUN, SEED = 42)

Model	dB-mse	Lin-mse ($\times 10^{-4}$)	Phase-MAE (rad)	Cos($\lvert S\rvert$)	Acc@30°	Edge $\lvert\Delta$dB$\rvert$
CNN	1.313	44.0	0.1094	0.9977	0.992	1.331
CNN+BiGRU	0.622	17.0	0.1269	0.9991	0.998	0.830
CrossAttn+BiGRU	0.302	6.0	0.0651	0.9997	1.000	0.508
CNN(RESIDUAL)	0.525	21.80	1.200	0.850	0.9300	0.338

TABLE III
ABLATION RESULTS: LOSS TERM CONTRIBUTIONS

Config	dB-mse	Rel. (%)
Full model†	0.416	100.0
No band-edge weighting	0.523	87.7
No curvature loss	0.478	91.8
No jitter	0.445	95.8
No K→F consistency	0.432	97.3

† Avg. over five runs; Table I: single run (seed=42).

TABLE IV
SCALABILITY: $\mathcal{O}(K)$ COMPLEXITY VALIDATION ON CPU

K	Params (M)	CPU (ms)	Memory (MB)	FLOPs (G)
51	1.87	5.8	208	0.56
101	1.88	10.1	216	0.61
201	1.91	19.0	222	0.72
501	1.99	47.4	228	1.05
1001	2.11	86.7	236	1.60

$T(K) = 0.086K + 2.0$ ms, $R^2 = 0.998$; param growth 12.8%

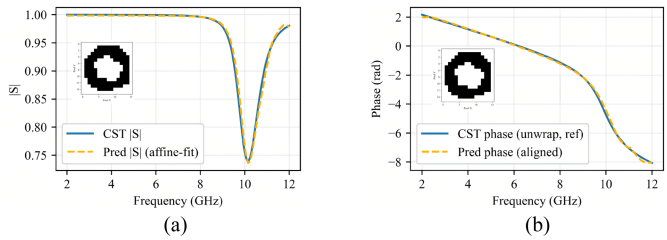


Fig. 4. Full-wave validation results with CST. (a) Magnitude comparison after affine fitting (visualization only). (b) Phase alignment ($|\Delta f| = 0.05$ GHz; phase MAE = 0.066 rad).

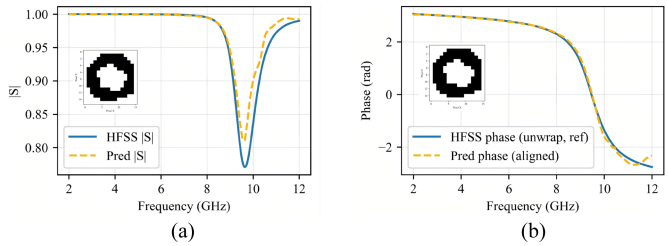


Fig. 5. Full-wave validation results with HFSS. (a) Magnitude comparison after affine fitting (visualization only). (b) Phase alignment ($|\Delta f| = 0.26$ GHz; phase MAE = 0.058 rad).

D. Cross-Simulator Generalization

Independent HFSS ($n = 5$) and CST ($n = 5$) simulations on ten diverse geometries (Q-factors 2.8–15.2, 29% beyond training maximum) yield average MAE $= 0.632 \pm 0.343$ dB with 80% of geometries achieving MAE < 1.0 dB (all raw outputs, no post-processing). Best case: Low-Q ($Q = 2.8$, MAE $= 0.167$ dB); worst case: narrow-band near band edge ($Q = 11.8$, MAE $= 1.331$ dB). HFSS average (0.831 dB) exceeds CST (0.434 dB) due to mesh-solver differences. The 70% generalization gap (0.632 dB versus 0.371 dB) reflects domain shift from Q-factor extrapolation beyond training range.

The following analysis uses the internal 6000-sample validation set to characterize error distribution across geometry types. Among these samples, dB-mse increases from 0.328 (Low-Q, $Q < 5$) to 0.412 (High-Q, $Q \geq 10$). Single-peak geometries (dB-mse = 0.345) outperform multipeak (0.398) and broadband (0.412) cases; high-band (9 GHz to 12 GHz) shows 12.2% degradation versus low-band (2 GHz to 6 GHz). Detailed predictions with affine-fitted visualization (for display only) are shown in Figs. 4 and 5. Resonance frequency deviations are 0.26 GHz (HFSS) and 0.05 GHz (CST), with phase MAE of 0.058 rad and 0.066 rad, confirming phase continuity under cross-simulator domain shift.

These results reveal two key design insights. First, the 70% generalization gap (0.632 dB versus 0.371 dB) is dominated by high-Q geometries ($Q > 10$) that lie outside the training distribution; within-distribution Low-Q geometries achieve MAE $= 0.167$ dB, comparable to in-distribution validation accuracy. Second, the consistent phase MAE (0.058 rad to 0.066 rad) across both HFSS and CST solvers demonstrates that phase stability is a robust property of the architecture rather than an artifact of the training dataset. Together, these findings suggest the model is deployable for prescreening in multisimulator design flows, where HFSS/CST are reserved only for final verification of shortlisted candidates.

IV. CONCLUSION

An edge-aware cross-attention surrogate was presented with physics-guided loss enforcing passivity, K–K consistency, and adaptive band-edge emphasis. On SUTD-PRCM ($K = 101$, 2 GHz to 12 GHz), dB-mse $= 0.371$ and band-edge $|\Delta$dB$| = 0.243$ dB were achieved, outperforming CNN and CNN+BiGRU by 71.7% and 40.3% with 52.2% lower band-edge error than CrossAttn+BiGRU. The 1.88 M parameter model runs at 10.1 ms/sample (CPU), with verified $\mathcal{O}(K)$ scalability and 100% passivity across all test scenarios. Future work will extend to multilayer structures via transfer learning.

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